

LINUX / OS EXAM ANSWERS

Q.3 (A) 1) What is Shell?

A Shell is a command-line interpreter in Linux/Unix that allows users to interact with the OS. It acts as a bridge between user and kernel. Functions: • Executes commands • Runs programs • Manages files • Supports scripting Example: bash

2) Variables statement (True/False)

This statement is FALSE. Special variables: \$? → Exit status \$\$ → Process ID \$# → Argument count \$0 → Script name \$1, \$2 → Arguments

3) Full form of GUI

GUI = Graphical User Interface

4) Full form of KDE

KDE = K Desktop Environment

Q.3 (B) 1) echo command

echo is used to display text or variables.

```
echo Hello
echo $HOME
```

2) passwd command

Used to change password.

```
passwd username
```

Q.3 (C) 1) break statement

Break stops loop immediately.

```
for i in 1 2 3 4 5
do
  if [ $i -eq 3 ]
  then
    break
  fi
  echo $i
done
```

2) continue statement

Continue skips current iteration.

```
for i in 1 2 3 4 5
```

```
do
  if [ $i -eq 3 ]
  then
    continue
  fi
  echo $i
done
```

Q.3 (D) 1) Odd Even Script

```
echo "Enter number:"
read num

if [ $((num % 2)) -eq 0 ]
then
  echo "Even"
else
  echo "Odd"
fi
```

2) Case Statement

```
echo "Enter number:"
read n

case $n in
  1) echo "One" ;;
  2) echo "Two" ;;
  *) echo "Invalid" ;;
esac
```

Q.4 (A) 1) GNOME

GNU Network Object Model Environment

2) GNOME Login

Username, Password, Session, Power options

3) Create Folder in Ubuntu

Use GUI or command:

```
mkdir foldername
```

4) Window Manager

Controls windows: open, close, move, resize

Q.5 (A) Full Forms

LDAP = Lightweight Directory Access Protocol
 DNS = Domain Name System
 FTP = File Transfer Protocol
 SMB = Server Message Block

Q.5 (B) WINS

Resolves NetBIOS name to IP

Web Server

Serves web pages (Apache, Nginx)

Q.5 (C) DNS

Converts domain to IP (www.google.com → IP)

Firewall

Protects network from unauthorized access

Q.5 (D) Samba

```
sudo apt install samba  
sudo systemctl start smbd
```

Active Directory

Used for: • User management • Security • Central control